1) What is cipher text. What is the difference between substitution and transposition cipher? State the differences between Vigenere cipher and ceaser cipher.

In cryptography, cipher text or cypher text is the result of encryption performed on plaintext using an algorithm, called a cipher. Cipher text is also known as encrypted or encoded information because it contains a form of the original plaintext that is unreadable by a human or computer without the proper cipher to decrypt it.

BASIS FOR COMPARISON	SUBSTITUTION TECHNIQUE	TRANSPOSITION TECHNIQUE
Basic	Replaces the plaintext characters with other characters, numbers and symbols.	Rearranges the position of the characters of the plaintext.
Forms	Monoalphabetic and polyalphabetic substitution cipher.	Keyless and keyed transpositional cipher.
Alterations	The identity of the character is changed while its position remains unchanged.	The position of the character is changed in spite of its identity.
Demerit	The letter with the low frequency can discern the plaintext.	Keys near to the correct key can disclose the plaintext.
Example	Caesar Cipher	Reil Fence Cipher

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BASIS FOR COMPARISON	VIGENERE CIPHER	CEASER CIPHER
Definition	A method of encrypting messages by using a series of different Ceaser cipher based on the letters of a particular keyword.	A type of substitution cipher in which each letter in the plaintext is 'shifted' a certain number of places down the alphabet
Strength	More powerful and much harder to crack compared to a single Ceaser cipher	Easier to crack and less powerful
Process	Uses rows of alphabets in the form of a square	Only replaces a letter of the alphabet with another letter that has been shifted
Feature	Polyalphabetic	Monoalphabetic
Decryption method can	Simple frequency method cannot be used to decrypt it	Simple frequency be used to decrypt it

2) Briefly explain DES and RSA algorithm with example.

RSA is an algorithm used by modern computers to encrypt and decrypt messages. It is an asymmetric cryptographic algorithm.Asymmetric means that there are two different keys.This is also called public key cryptography, because one of the keys can be given to anyone. The other key must be kept private.
The keys for the RSA algorithm are generated the following way:

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Key Generation.

1. Select two prime numbers p & q.
2. Calculate n = pxq.
3. Calculate $\Phi(n) = (p-1)x(q-1)$
4. Choose a value for e.

5. Calculate $d = e^{-1} \bmod \Phi(n)$
6. Public key PU = {e,n}
7. Private key PR = {d,n}

So PU = {7,33} & PR = {3,33}

Let p = 3 , q= 11
$n = p \times q = 3 \times 11 = 33$
So n = 33.
$\Phi(n) = (3-1)x(11-1) = 2x10 = 20.$
So $\Phi(n) = 20.$

So let e = 7
$1<7<20$ & $\gcd(7,20) = 1$

$d = e^{-1} \bmod \Phi(n)$
i.e, $ed = 1 \bmod \Phi(n)$
ie, $7 \times d = 1 \bmod 20$

$7 \times 3 \bmod 20 = 21 \bmod 20 = 1$
so d = 3

Encryption.

Plaintext , $M < n$

$$C = M^e \bmod n$$

Let M be 31.
Then $C = 31^7 \bmod 33$
$= 4.$

So PU = {7,33} & PR = {3,33}

p = 3 , q= 11

n = 33.

$\Phi(n) = 20.$

e = 7

d = 3

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Decryption.

Ciphertext . C

$M = C^d \bmod n$

So C = 4.

Then $M = 4^3 \bmod 33$

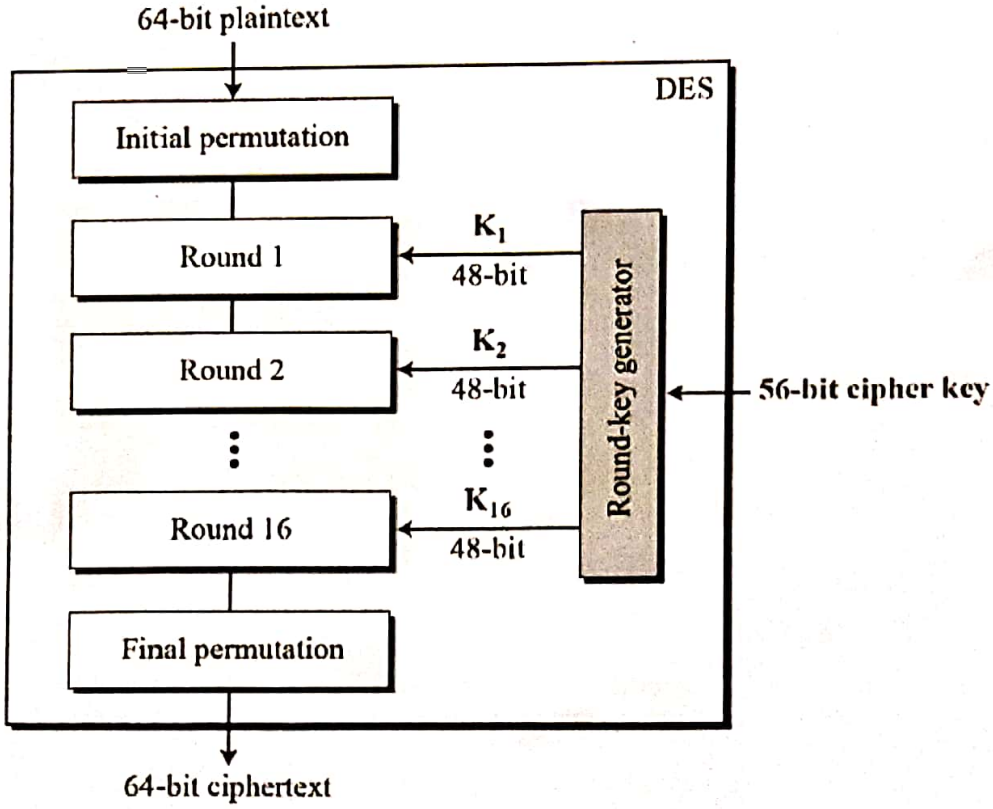
$= 31$..

So PU = {7,33} & PR = {3,33}

DES is an implementation of a Feistel Cipher. It uses

- 16 round Feistel structure.
- 64 bit block data
- 56 bit key
- Single key for encryption and decryption

General Structure of DES is depicted in the following illustration –



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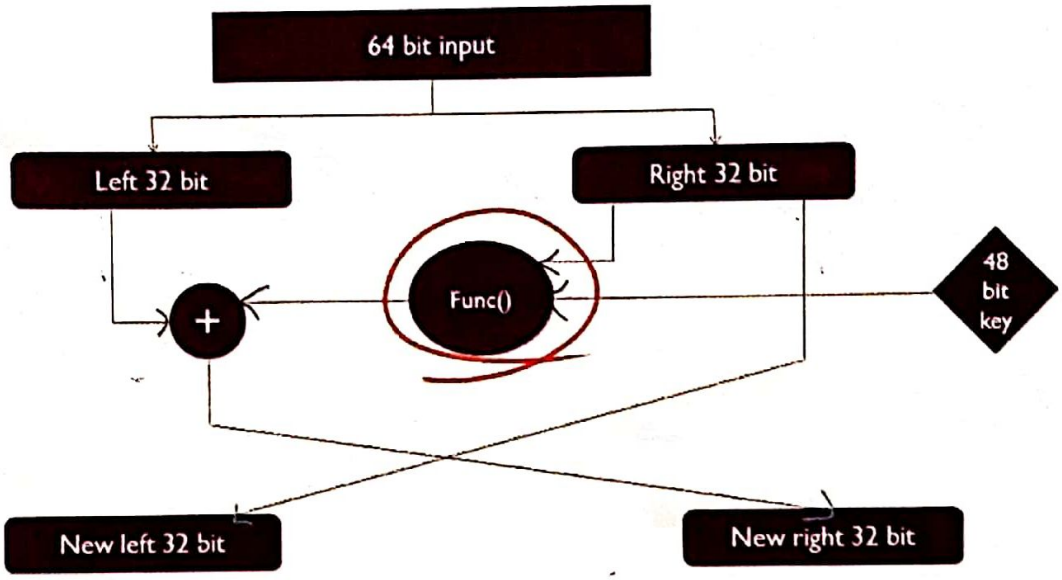
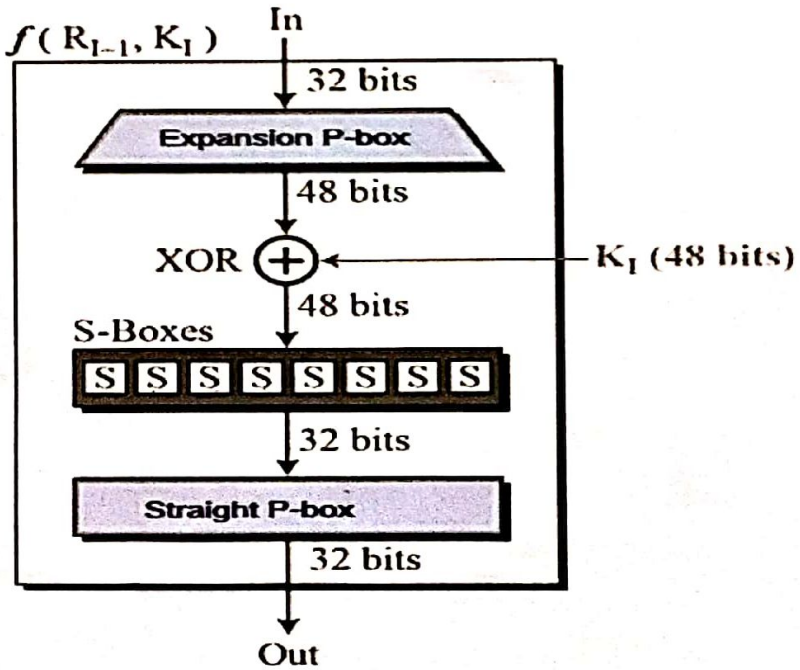
DATA ENCRYPTION STANDARD (DES) - ONE ROUND



The above figure shows the details of each rounds. In each round, each stage takes two 32-bit inputs and produces two 32-bit outputs. The left output is simply a copy of the right input. The right output is the bitwise XOR of the left input and a function of the right input and the key for this stage. All the complexity lies in this function. The DES function applies a 48-bit key to the rightmost 32 bits to produce a 32-bit output.



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3) Distinguish between symmetric and asymmetric key cryptography.

BASIS FOR COMPARISON	SYMMETRIC ENCRYPTION	ASYMMETRIC ENCRYPTION
Basic	Symmetric encryption uses a single key for both encryption and Decryption.	Asymmetric encryption uses a different key for encryption and decryption.
Performance	Symmetric encryption is fast in execution.	Asymmetric Encryption is slow in execution due to the high computational burden.
Algorithms	DES, 3DES, AES, and RC4.	Diffie-Hellman, RSA.
Purpose	The symmetric encryption is used for bulk data transmission.	The asymmetric encryption is often used for securely exchanging secret keys.

4) A key of transposition cipher is 'question'. Plaintext is 'we love our country bangladesh'. Find the encrypted cipher text.

There are 8 letters in 'question' so we start by making rows of 8 columns. There are 26 letters in total. So we need 4 rows of 8. If necessary pad the message with a's.

1	2	3	4	5	6	7	8
w	e	l	o	v	e	o	u
r	c	o	u	n	t	r	y
b	a	n	g	l	a	d	e
s	h	a	a	a	a	a	a

Now using the key 'question' we order the columns: 3 6 8 7 1 4 5 2

So the cipher text will be: lonaetaauyeaordawrbsougavnlaecah

5) Using RSA algorithm encrypt the word 'computer'.

Using the RSA encryption method we get the following steps:

1) Choose two prime numbers $p = 17$ and $q = 11$
2) Compute $n = pq$ and $z = (p - 1)(q - 1)$

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a. n = pq = (17)(11) =187
b. z = (p – 1)(q – 1) = (17– 1)(11 – 1) = (16)(10) = 160
3) Choose a number e < n such that it has no common factors with z other than 1
a. Let e = 7
4) Find a number d such that ed= 1 mod z
a. ed= 1 mod 161
d= 23
5) The public key becomes the number pair (n, e) and the private key becomes the number pair (n, d) Thus, the encrypted value c of the plain text message m is: c = m^e mod n Now, using the alphabet such that the letters are numbered 1 through 26 we get for 'computer':

Plaintext:	C	O	M	P	U	T	E	R
Number:	3	15	13	16	21	20	5	18
Encrypted value:	130	93	106	135	98	147	146	171

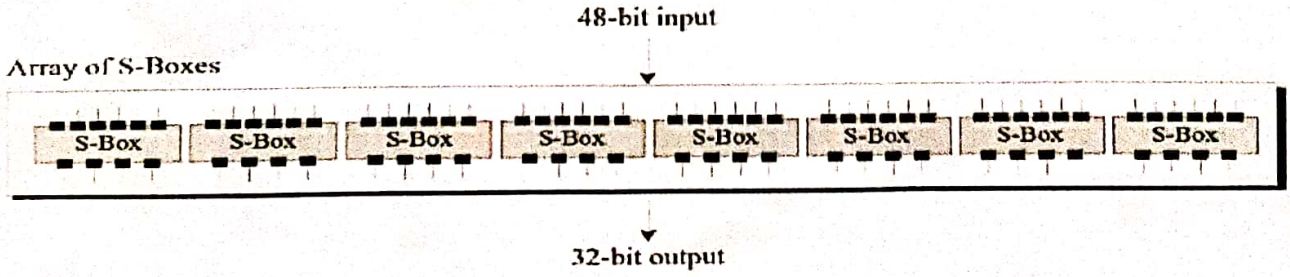
Where,
3^7 mod 187=130
15^7 mod 187=93
13^7 mod 187=106
16^7 mod 187=135
21^7 mod 187=98
20^7 mod 187=147
5^7 mod 187=146
18^7 mod 187=171

6) Why do we use P-box and S-box in DES? Explain with example.

S-box: In cryptography, an S-box (substitution-box) is a basic component of symmetric key algorithms which performs substitution. In block ciphers, they are typically used to obscure the relationship between the key and the ciphertext — Shannon's property of confusion.

In general, an S-box takes some number of input bits, *m*, and transforms them into some number of output bits, *n*, where *n* is not necessarily equal to *m*. An $m \times n$ S-box can be implemented as a lookup table with $2m$ words of *n* bits each. Fixed tables are normally used, as in the Data Encryption Standard (DES), but in some ciphers the tables are generated dynamically from the key.

DES uses 8 S-boxes, each with a 6-bit input and a 4-bit output. Refer the following illustration –



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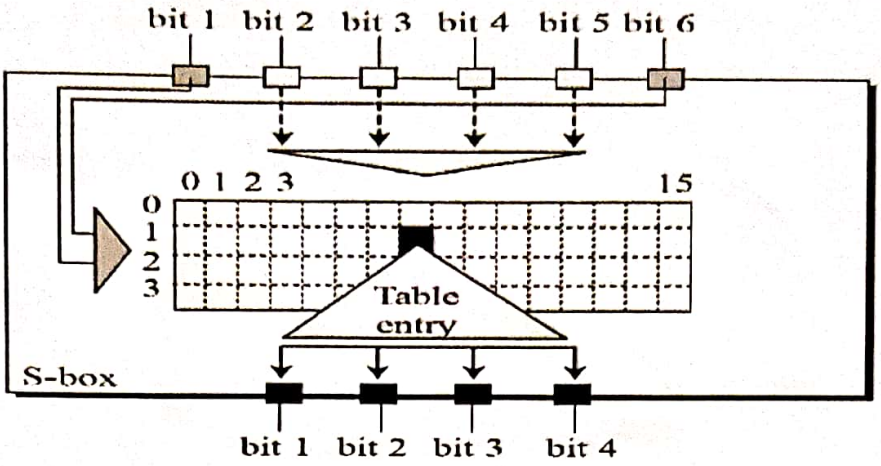
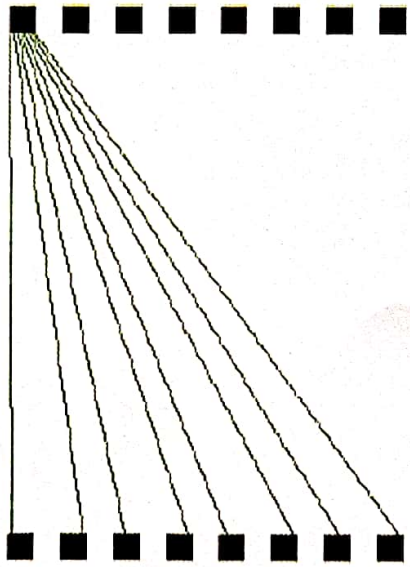


P-box: In cryptography, a permutation box (or P-box) is a method of bit-shuffling used to permute or transpose bits across S-boxes inputs, retaining diffusion while transposing. P-boxes are typically classified as *compression*, *expansion*, and *straight*, depending on whether the number of output bits is less than, greater than, or equal to the number of input bits. Only straight P-boxes are invertible.



An example of a 64-bit P-box which spreads the input S-boxes to as many output S-boxes as possible.

7) Define digital signature and Steps of Diffie-hillman algorithm.

A digital signature is a mathematical technique used to validate the authenticity and integrity of a message, software or digital document.

The Diffie Hillman algorithm is illustrated:

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- Prerequisite: N is a large prime number such that (N-1)/2 is also a prime number. G is also a prime number. Both N and G are known to Ram and Sita..
- Sita chooses a large random number x and calculates $R1 = G^x \bmod N$ and sends it to Ram
- Ram chooses another large random number y and calculates $R2 = G^y \bmod N$ and sends it to Sita
- Ram calculates $K = (R1)^y \bmod N$
- Sita calculates $K = (R2)^x \bmod N$

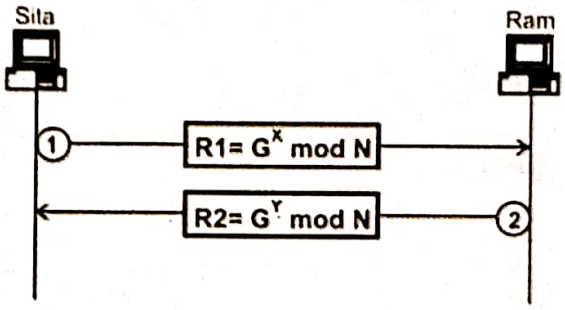

Figure 8.2.11 Diffie-Hellman Protocol

Example:

Ans: Although the algorithm works on large numbers, it is illustrated with smaller numbers in this example.
Let N = 23 and G = 7.
Sita chooses x = 5 and calculates $R1 = 7^5 \text{Mod} 23 = 17$
Sita sends 17 to Ram.
Ram chooses y = 3 and calculates $R2 = 7^3 \text{Mod} 23 = 21$
Ram sends 21 to Sita
Ram calculates $K = 17^3 \text{Mod } 23 = 14$
Sita Calculates $K = 21^5 \text{Mod} 23 = 14$

8) What do you mean by ACL. Why do you use ACL in networking.

ACL stands for Access Control List. It is a set of rules defined for controlling network traffic and reducing network attack. ACLs are used to filter traffic based on the set of rules defined for the incoming or outgoing of the network.

There are a variety of reasons we use ACLs. They are listed below:

- To provide a basic level of security for the network.
- ACLs are not as complex and in depth of protection as firewalls, but they do provide protection on higher speed interfaces where line rate speed is important and firewalls may be restrictive.

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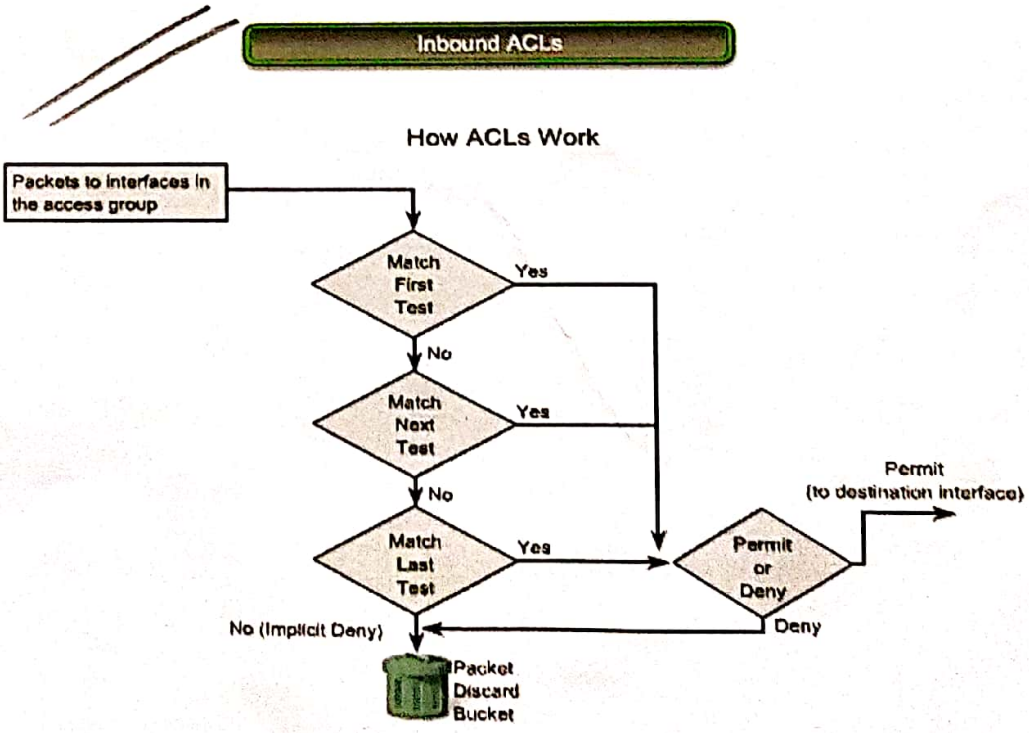
- To restrict updates for routing from network peers
- Can be instrumental in defining flow control for network traffic.
- ACLs placed on external routes to filter traffic against less desirable networks
- Denial of services is controlled

9) Explain operations of inbound and outbound ACL (difference + flowchart)

Any packet coming to the router is considered as inbound. Any packet going out of the router is considered as outbound. In ACL, we usually use it at interface, any packet entering to interface is considered as inbound by ACL, any packet leaving out of interface is considered as outbound by ACL.

Inbound ACL	Outbound ACL
When an access list is applied on inbound packets of the interface then first the packets will processed according to the access list and then routed to the outbound interface.	When an access list is applied on outbound packets of the interface then first the packet will be routed and then processed at the outbound interface.
Creates more router load	Less router load is created compared to inbound
More efficient	Less efficient
Any matched 'deny' packet is dropped before the packet gets routed to the destined outbound interface	The packets get rooted to the outbound interface and then denied packets are dropped



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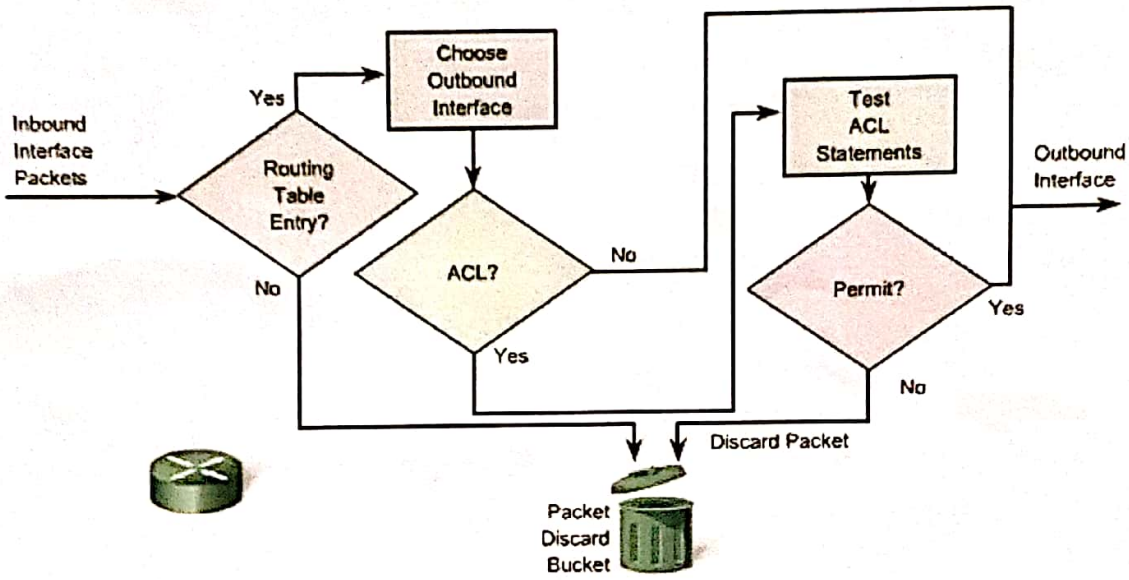
Outbound ACLs

Outbound ACL Example



10) How implicit deny use in ACL. Distinguish between standard and extended ACL, wildcard mask and subnet mask.

An implicit deny is when a user or group are not granted a specific permission in the security settings of an object, but they are not explicitly denied either.

By default, there is an implicit deny all clause at the end of every ACL. Anything that is not explicitly permitted is denied. Note: The ACL access-list 1 permit 192.168.10.1 0.0.0.0 is another way to configure the same rule.

Standard ACL	Extended ACL
Allow filtering based on source address.	Allow filtering based on source and destination addresses, as well as protocol and port number.
Used to block particular host, network or subnet.	Used to block particular services.
Standard ACL is implemented close to the destination	Extended ACL is implemented closet to the source.
Standard ACL Range–> 1 – 99 & 1300-1999.	Extended ACL Range–> 100 – 199 & 2000 – 2699

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In Standard ACL, two-way communications will be blocked	In Extended ACL, one-way communication will be blocked
In Standard ACL, all services will be blocked	Extended ACL, particular services will be blocked.

Subnet Mask	Wildcard Mask
A 32 bit value which differentiates the host portion and network portion of an IP address	Defines which IP addresses are allowed and which are blocked
Network portion is designated by 1's and host portion by 0's	Accurate matches are defined by 0's while any non -zero values define any value on the corresponding octet
Used when we configure any IP address or an interface	Used when we configure networks in ACL
Less flexible	More flexible

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